

PERSONA

Management Software

1. INTRODUCTION

In the contemporary digital era, organizations face the intricate challenge of managing a diverse array of user profiles—including employees, vendors, customers, and others—across multiple platforms. This fragmentation often results in data silos, operational inefficiencies, and security vulnerabilities. For instance, a report by Mercer Bucks Technology highlights that 60% of organizations encounter difficulties in managing user access as they scale . Furthermore, the IT Services and Technology Association found that implementing centralized user management can reduce administrative task time by 30%, underscoring the operational benefits of such systems .

Recognizing these challenges, our team embarked on the development of the **Persona Module**—a comprehensive system designed to streamline the categorization and management of various personas within a unified dashboard. This initiative aims to enhance data accessibility, improve operational efficiency, and fortify security measures. By consolidating user information into a single platform, the Persona Module seeks to eliminate data silos, facilitate seamless access control, and ensure consistent application of security policies. This report delves into the objectives, features, and anticipated impact of the Persona Module, underscoring its significance in modern organizational contexts.

2. OBJECTIVE

The primary objective of the **Persona Module** project is to develop a centralized system that streamlines the management of diverse user profiles—including employees, vendors, customers, and others—within an organization. By consolidating user information into a unified platform, the project aims to enhance data accessibility, improve operational efficiency, and strengthen security protocols.

Key objectives include:

1. **Centralized User Management:** Establish a single, cohesive system that consolidates all user profiles, thereby reducing data fragmentation and eliminating silos. This centralization facilitates easier access to user information and simplifies administrative processes.
2. **Dynamic Persona Categorization:** Implement a flexible framework that allows for the dynamic categorization of users into specific personas—such as employees, vendors, customers, and others—based on their roles and interactions with the organization. This categorization enables tailored management and communication strategies for each user group.
3. **Enhanced Data Accuracy and Consistency:** Ensure that user information is consistently updated and maintained across the organization, reducing discrepancies and improving the reliability of data used in decision-making processes.
4. **Improved Security and Compliance:** Strengthen security measures by enforcing standardized access controls and permissions across all user profiles. Centralized

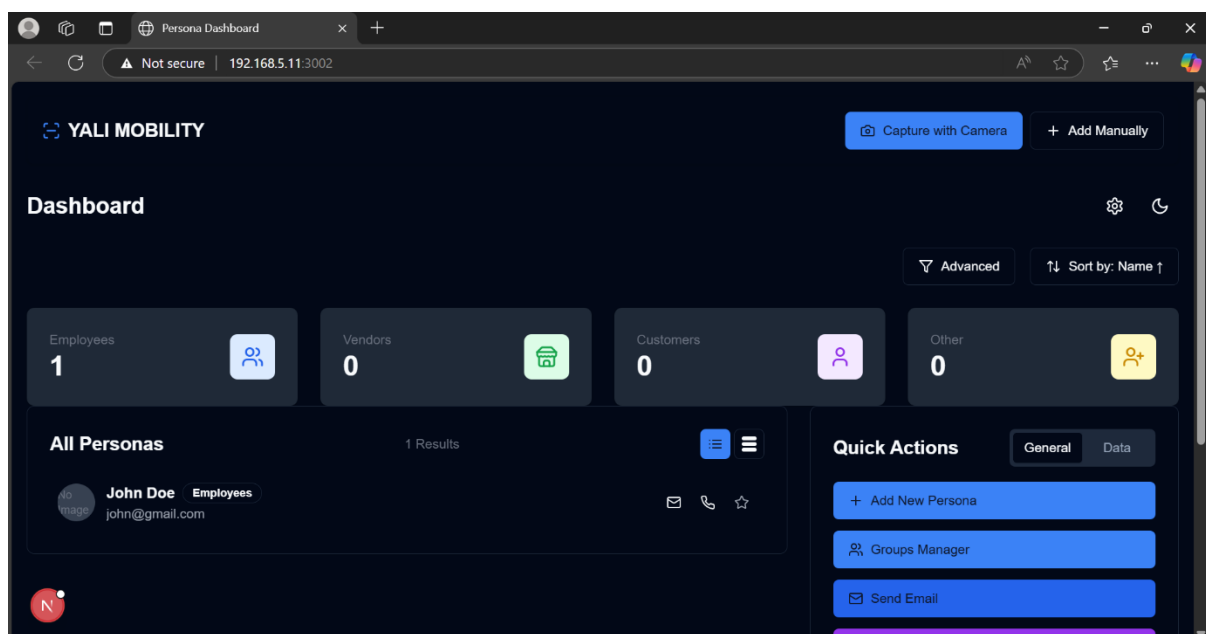
management allows for consistent application of security policies, aiding in compliance with regulatory standards and reducing the risk of unauthorized access.

5. **Operational Efficiency:** Streamline workflows related to user onboarding, role changes, and offboarding by providing a unified interface for managing user information. This efficiency reduces administrative overhead and accelerates response times to user-related requests.
6. **Scalability and Flexibility:** Design the system to accommodate organizational growth and evolving user management needs. The module should be adaptable to incorporate new user categories and integrate with emerging technologies as required.

3. PROJECT DESCRIPTION

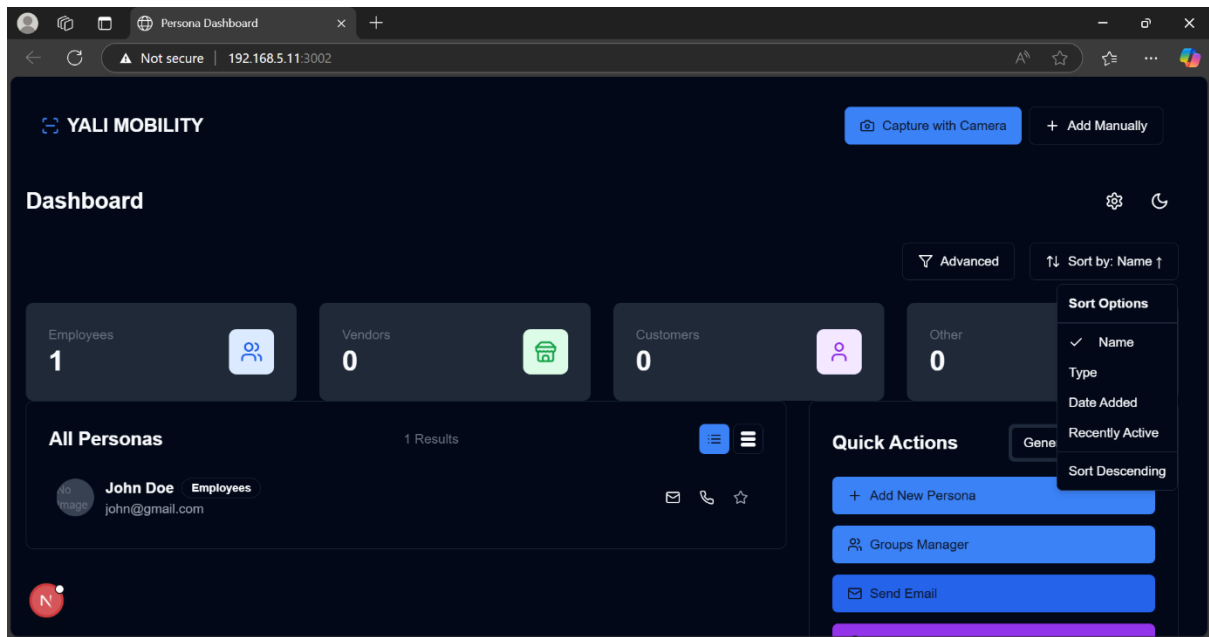
The **Persona Module Dashboard** serves as the central interface for managing user personas, offering a structured and efficient approach to overseeing employees, vendors, customers, and other users. It provides a clear overview of registered personas, displaying categorized sections with count indicators for quick insights. The dashboard simplifies persona management by allowing users to be added manually or captured using a camera, ensuring flexibility in data entry. With built-in search, sorting, and filtering options, retrieving specific personas becomes seamless, enhancing user accessibility.

A dedicated **Quick Actions** panel offers essential functions such as adding new personas, managing groups, and sending emails directly from the dashboard, streamlining communication and administrative tasks. Individual persona details, including name, email, and assigned category, are displayed for easy identification, along with quick-contact options like email and call integration. Designed with a sleek **dark mode interface**, the dashboard enhances readability and user experience while offering customization options for better workflow management. It serves as a comprehensive system for organizing, tracking, and engaging with different personas efficiently.



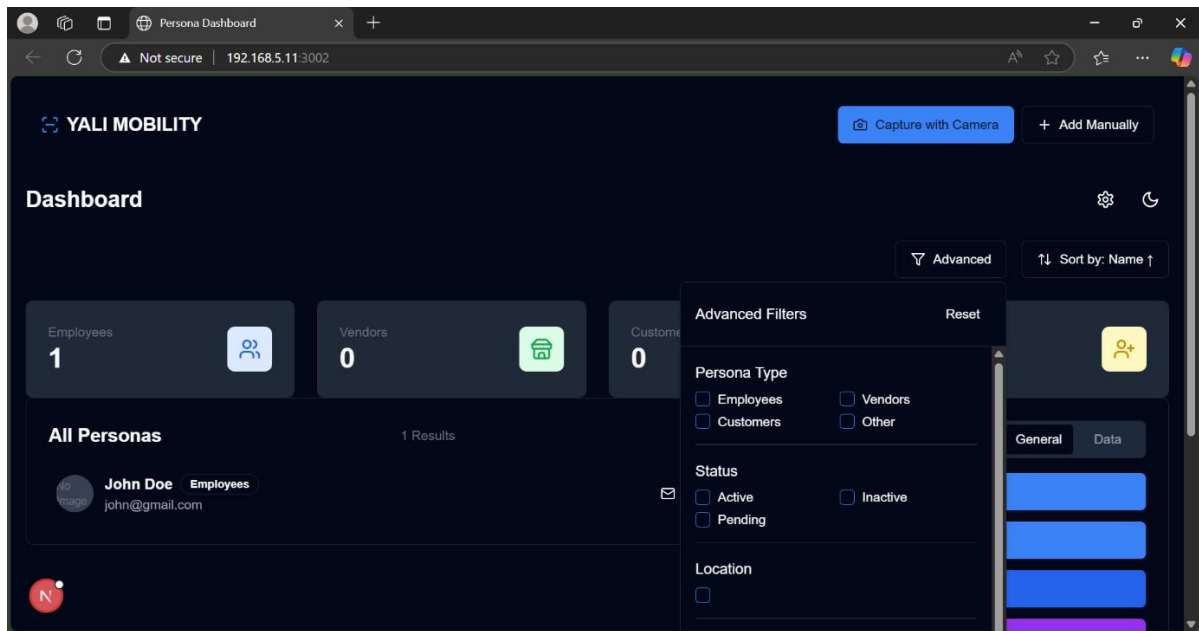
3.1 Sorting in Dashboard

The **Sorting Feature** in the Persona Module Dashboard enhances data organization by allowing users to arrange personas based on specific criteria. Users can sort by **Name, Type, Date Added, and Recently Active**, ensuring quick and efficient access to relevant personas. Additionally, the **Sort Descending** option provides flexibility in viewing data in reverse order. Integrated within an intuitive dropdown menu, this feature simplifies navigation and improves user efficiency by offering seamless control over data arrangement, making persona management more structured and accessible.



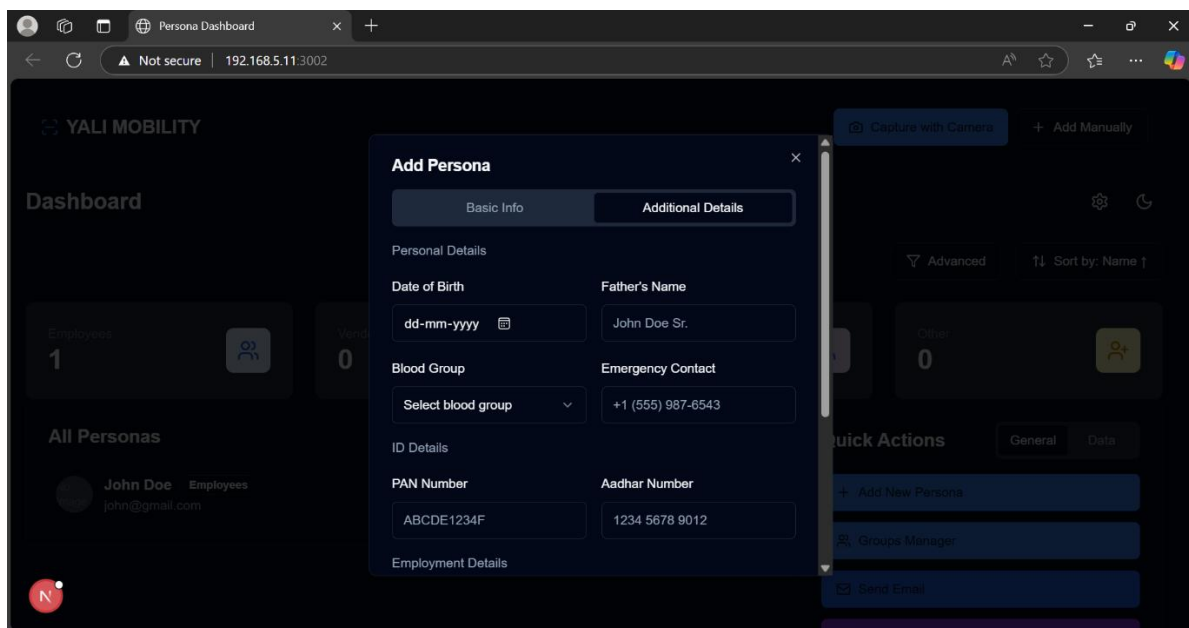
3.2 Advanced Filtering

The **Advanced Filtering** feature in the Persona Module Dashboard provides users with a powerful tool to refine search results and locate specific personas efficiently. Users can apply filters based on **Persona Type (Employees, Vendors, Customers, Other), Status (Active, Inactive, Pending), and Location**. The **Reset** option allows quick removal of all applied filters for a fresh search. Integrated seamlessly into the interface, this feature ensures an intuitive and user-friendly experience, making persona management more streamlined and tailored to user needs.



3.3 Add Manually

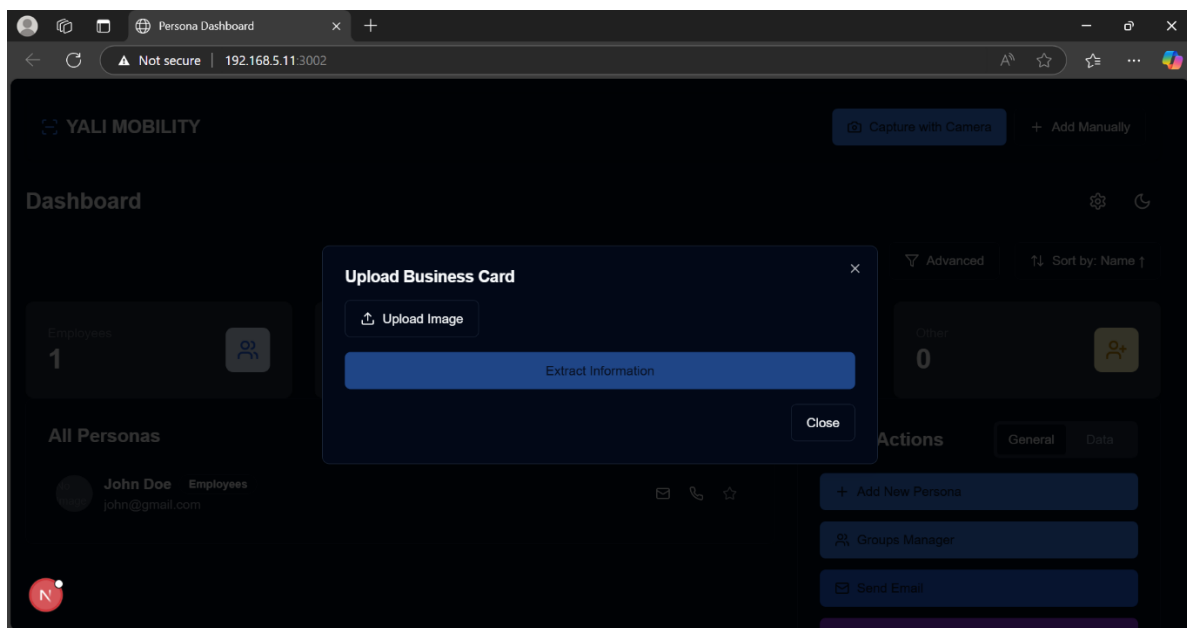
The **Add Manually** feature in the Persona Module Dashboard enables users to create new personas by entering details manually. This feature allows users to input **Basic Info (Name, Email, Persona Type)**, **Personal Details (Date of Birth, Father's Name, Blood Group, Emergency Contact)**, and **ID Details (PAN, Aadhar Number)** with ease. The structured form ensures data accuracy and completeness while providing a seamless user experience. Positioned alongside the **Capture with Camera** option, this feature offers flexibility in persona creation, making it highly efficient for manual data entry.



3.4 OCR (Business Card Upload)

The **OCR (Optical Character Recognition)** feature allows users to extract information from a business card by uploading its image. As shown in the interface, users can click on "**Upload Image**" to select a business card, and then tap on "**Extract Information**" to automatically populate fields like **Name, Email, Phone Number, Designation, and Company Name**.

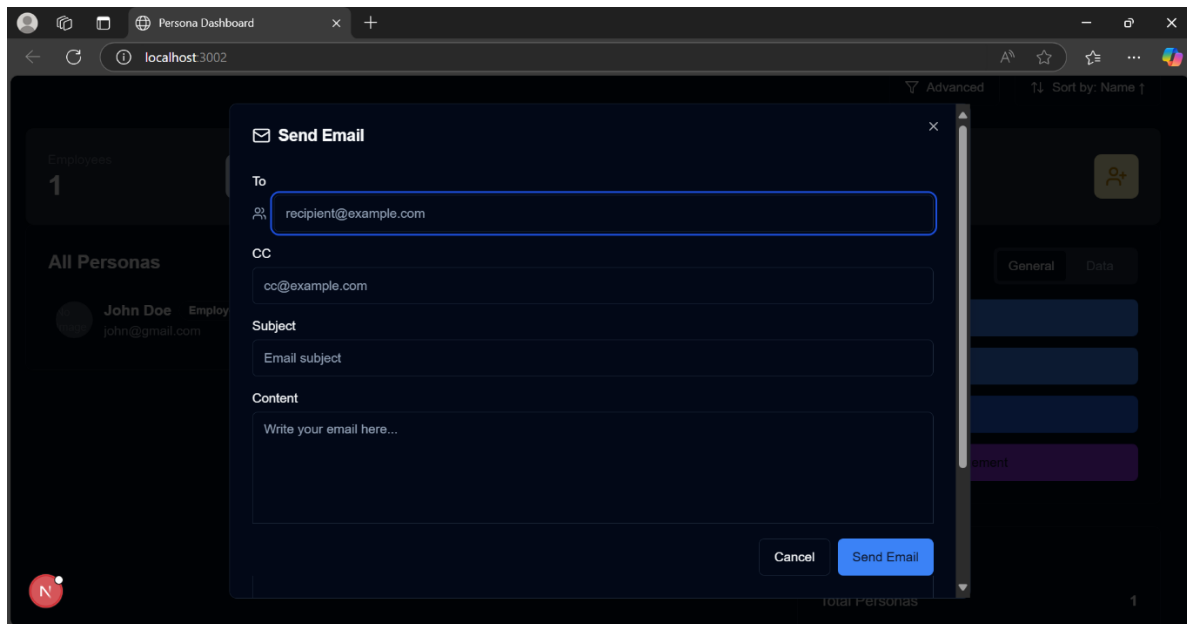
This functionality speeds up the data entry process and improves accuracy by leveraging image-to-text conversion, making it easier to create personas with minimal manual input. It is especially useful in professional settings where quick digitization of business card details is required.



3.5 Send Email (Quick Action)

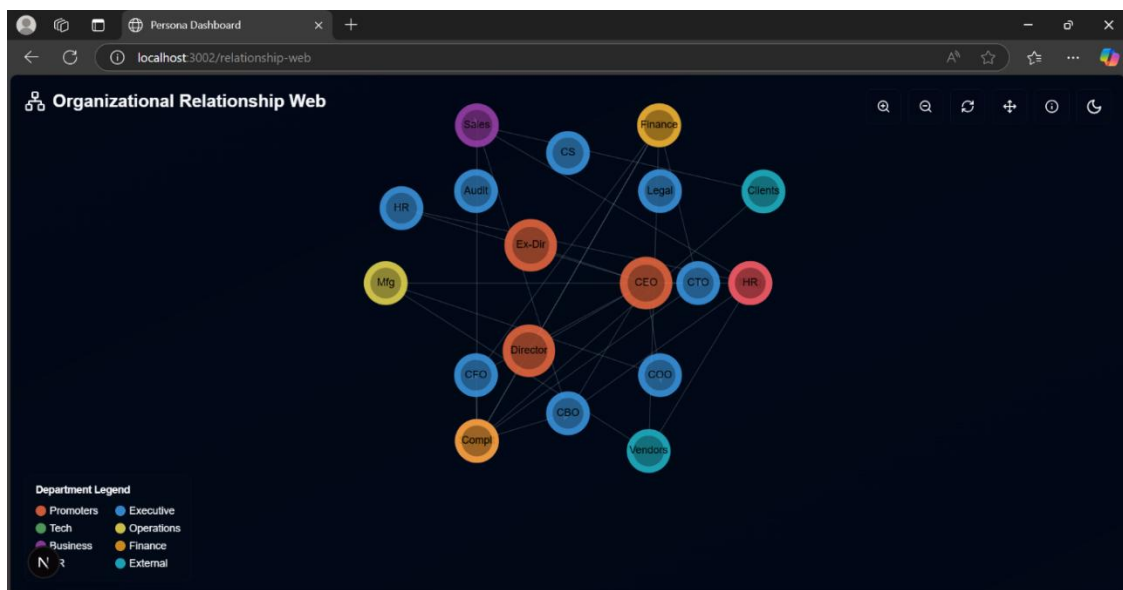
The **Send Email** feature under **Quick Actions** allows users to seamlessly compose and send emails directly from the dashboard. As shown in the interface, users can input the **recipient's email (To)**, add **CC recipients**, specify a **subject**, and type in the **content** of the message—all within a modal pop-up.

This feature simplifies communication by eliminating the need to switch between applications. With a clean UI and easy-to-access fields, it enables users to reach out to clients or team members instantly, enhancing productivity and streamlining workflow.



3.6 Organizational Relationship Web (Quick Action)

The **Organizational Relationship Web** feature offers a visually interactive representation of internal and external organizational relationships. Located under **Quick Actions**, this tool enables users to quickly understand the chain of command, interdepartmental connections, and external collaborations. As illustrated, each node represents a key role or department, categorized by color-coded **Department Legends** such as Promoters, Executive, Tech, Operations, Business, Finance, External, and HR. The web format makes it intuitive to track direct and indirect relationships, fostering better strategic decisions and improved team coordination. This feature is especially useful during meetings, audits, or strategic planning sessions, allowing stakeholders to gain instant clarity on organizational dynamics.

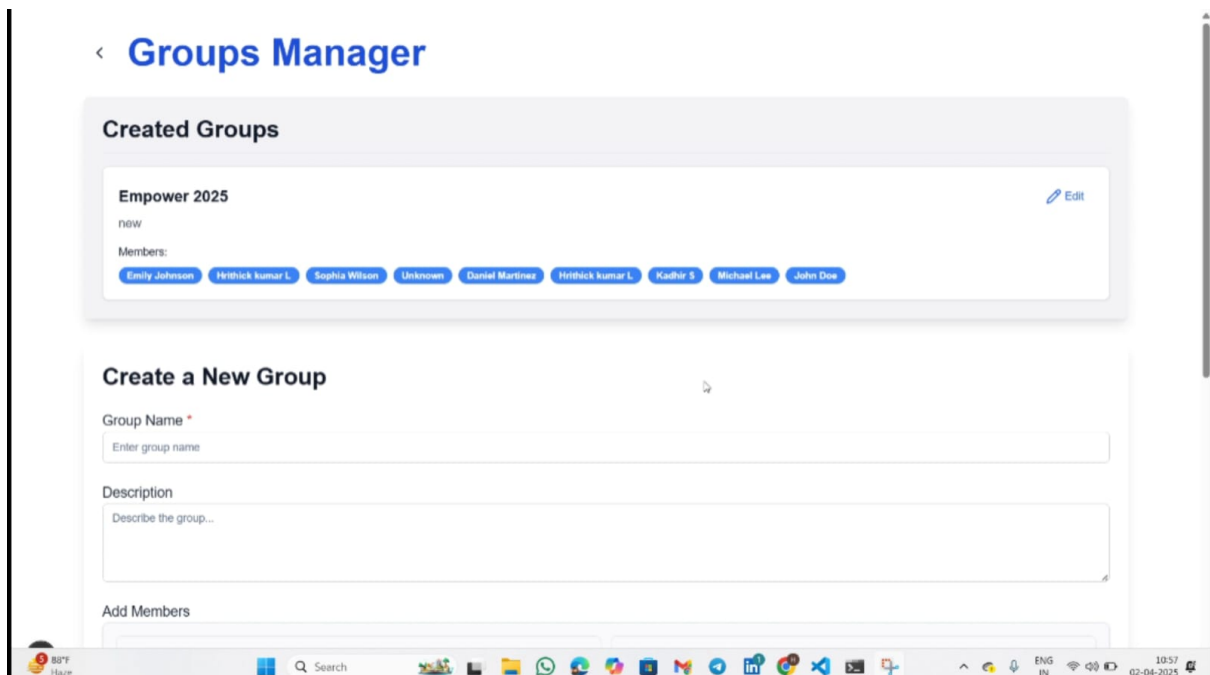


3.7 Group Manager (Quick Action)

The **Group Manager** under **Quick Action** empowers users to seamlessly **create, manage, and edit groups** within the organization. This feature is designed to improve collaboration by organizing individuals into focused teams.

As shown in the screenshot, a group named "**Empower 2025**" has been created, displaying its list of members through visually distinct tags. Users can easily add members like *Emily Johnson*, *Sophia Wilson*, and *Michael Lee*, or even edit existing groups via the intuitive **Edit** option.

The interface also allows users to **Create a New Group** by simply entering a **group name**, providing a **description**, and selecting team members. This functionality ensures quick setup and easy updates of team structures, making it ideal for managing project teams, committees, or initiative-based groups.



3.8 Analytics Dashboard (Quick Action)

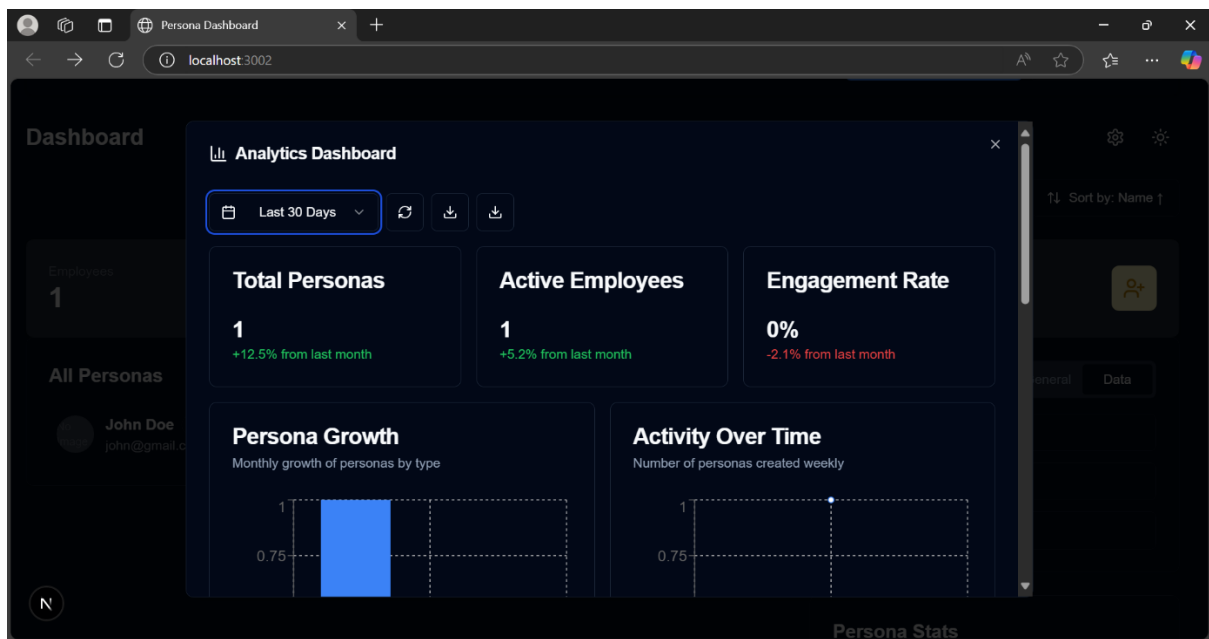
The **Analytics Dashboard** under **Quick Action** provides a comprehensive overview of persona activity and employee engagement across a selected timeframe.

As seen in the screenshot, users can filter analytics using a dropdown for "**Last 30 Days**" or refresh the data in real time. Key metrics include:

- **Total Personas:** Shows the total number of personas created (currently 1), along with a **+12.5%** growth from last month.

- **Active Employees:** Indicates the number of employees actively participating (currently 1), showing a +5.2% increase.
- **Engagement Rate:** Currently at 0%, indicating a -2.1% drop from the previous month.

Below the statistics, visual charts like **Persona Growth** and **Activity Over Time** help users track persona creation patterns and trends on a weekly basis. This dashboard ensures users have quick, actionable insights to evaluate participation and growth effectively.



3.9 Import Data (Quick Action)

The **Import Data** feature under **Quick Action** allows users to seamlessly upload persona information in bulk through a CSV file.

As seen in the interface, users can:

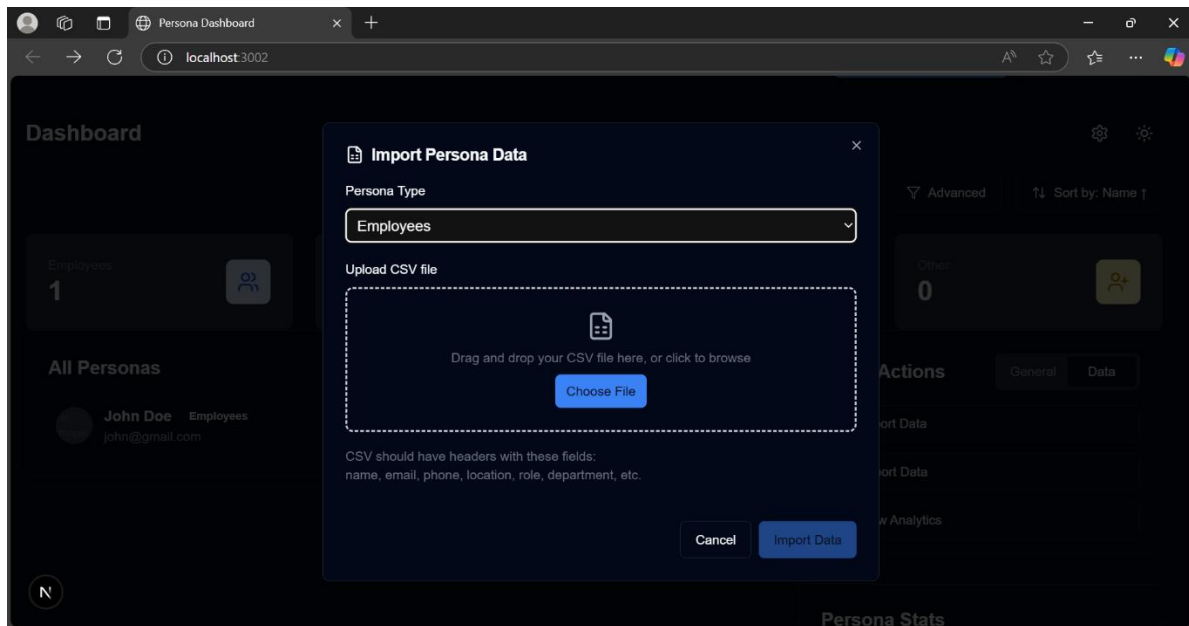
- **Select the Persona Type** (e.g., Employees) using a dropdown menu.
- **Upload CSV Files** by either dragging and dropping the file or clicking the “**Choose File**” button.

A clear instruction is provided that the CSV file should include headers such as:

name, email, phone, location, role, department, etc.

This ensures structured data entry for creating multiple personas efficiently. Once the file is uploaded, users can simply click the “**Import Data**” button to populate the dashboard with the new personas.

This feature greatly enhances scalability by eliminating the need for manual entry and supports faster onboarding of employee or user personas.

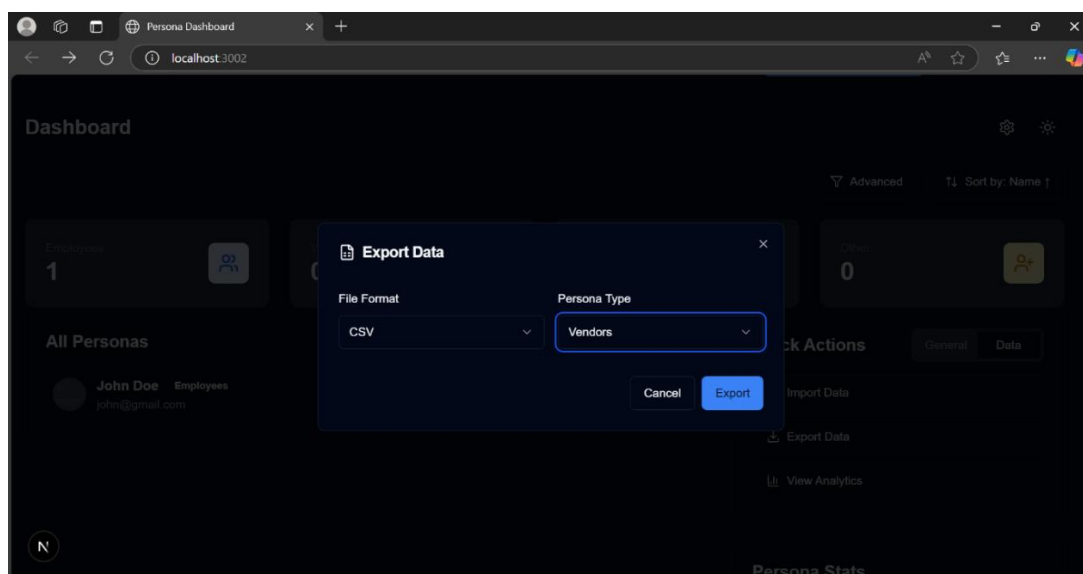


3.10 Export Data (Quick Action)

The **Export Data** feature under **Quick Action** allows users to extract persona data in a structured file format for external use or backup.

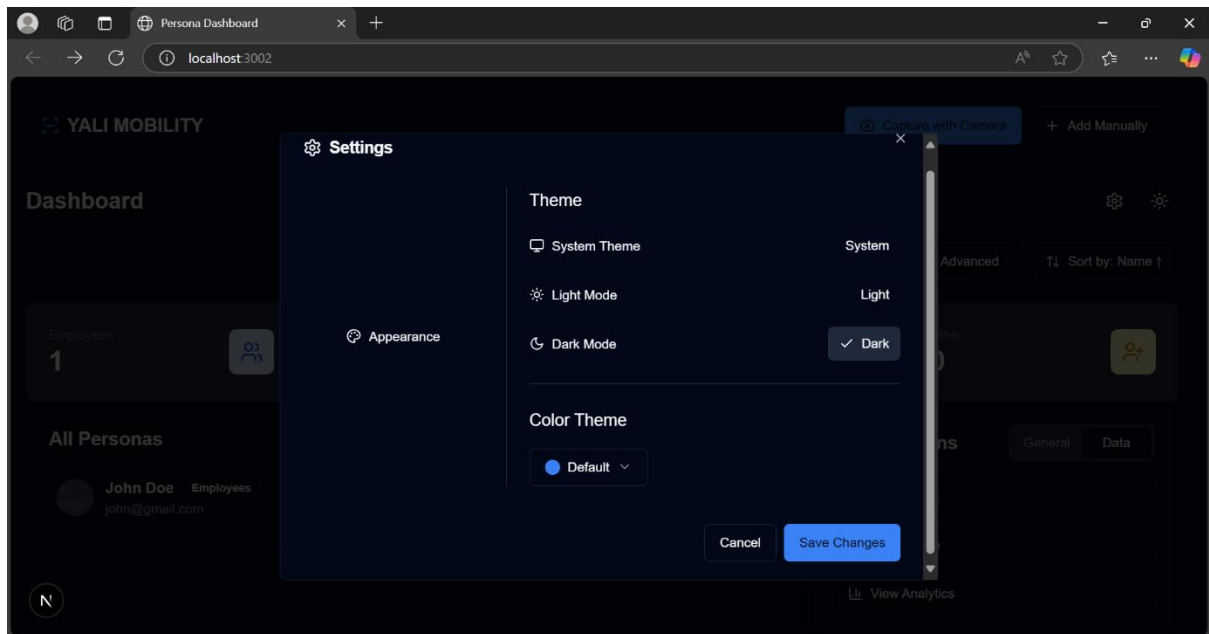
In the interface:

- Users can select the **File Format** (e.g., **CSV**) from a dropdown.
- They can also choose the specific **Persona Type** (e.g., **Vendors**) to export.
- Once selections are made, clicking the **“Export”** button triggers the download of the data file.
- A **“Cancel”** button is available to exit the action without exporting.



3.11 Settings Feature

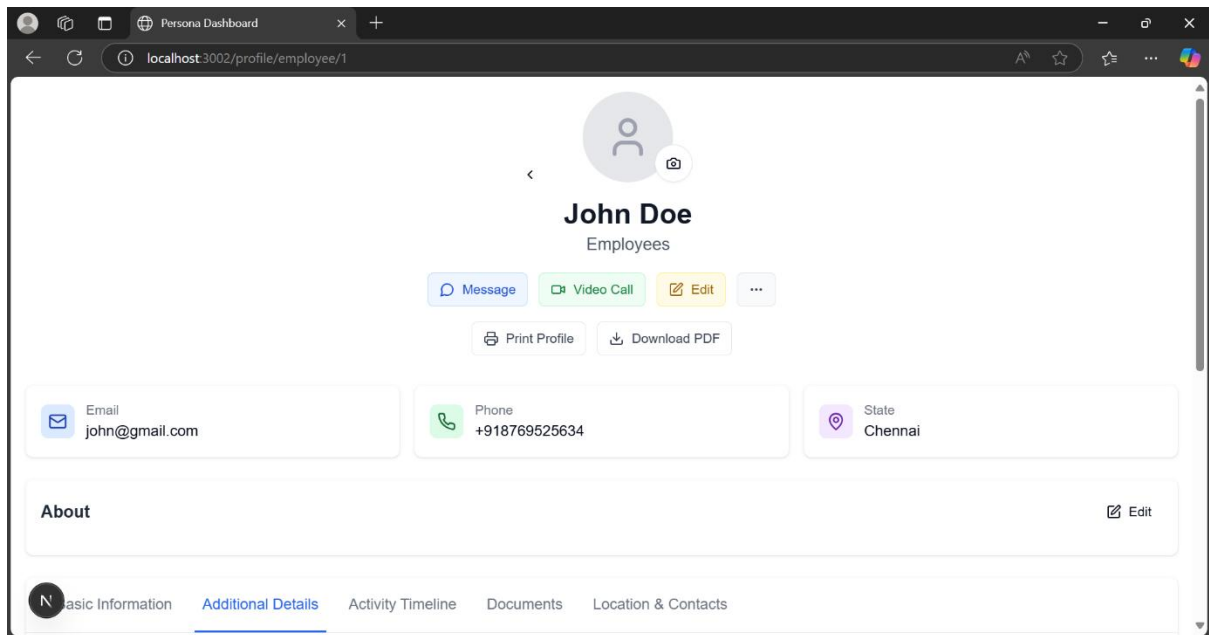
The Settings feature allows users to customize the dashboard by selecting from System, Light, or Dark themes and choosing a preferred color theme. I contributed to implementing the theme-switch functionality using state management, ensuring smooth and instant UI updates. I also tested the feature across different devices for consistency. This feature enhances user experience by offering personalization and improving visual comfort.



3.12 Profile Page

The Profile Page displays detailed information about each persona, such as their name, type (e.g., Employee), contact details, and location. It includes options to message, video call, edit details, print the profile, or download it as a PDF. The page is organized into tabs like Basic Information, Additional Details, Activity Timeline, Documents, and Location & Contacts for easy navigation.

I contributed by ensuring smooth layout rendering, organizing information into tabs, and testing interactive features like editing and PDF export. This page improves data visibility and enhances communication with individual personas.

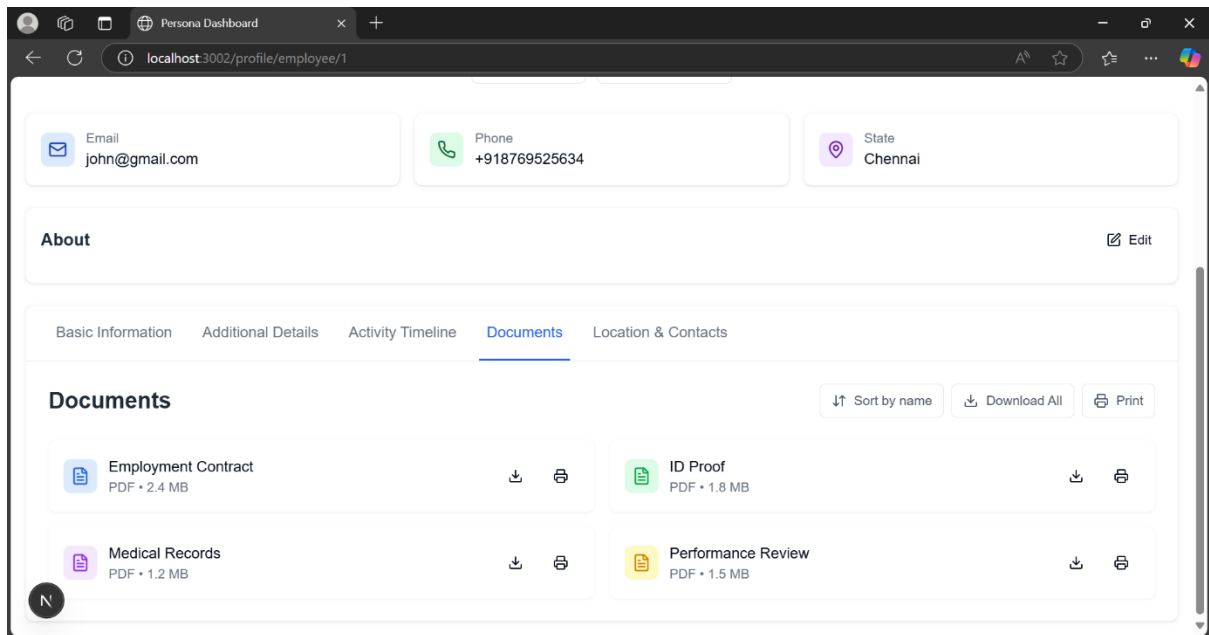


3.13 Additional Tabs: Documents, Timeline & More

The profile page includes multiple useful tabs like **Documents**, **Activity Timeline**, **Additional Details**, **Basic Information**, and **Location & Contacts**. In the **Documents** tab, we can upload and manage important files like ID Proofs, Medical Records, Employment Contracts, and more. Each document can be downloaded or printed individually, and users also have the option to download or print all documents at once.

The **Activity Timeline** tab captures key milestones and updates related to the persona, such as their date of birth, joining date, and performance events. This feature helps in tracking personal and professional highlights in a well-organized manner.

I worked on testing the document upload feature and ensuring file previews and actions like download and print worked smoothly. For the timeline, I verified date handling and visibility for different events, contributing to a structured and informative user experience.



4.USER MANUAL

4.1 Overview

The **Persona Module** is a comprehensive digital tool designed to streamline the management of various user profiles within an organization. It supports the handling of different persona types such as **Employees**, **Vendors**, **Customers**, and **Others**. The module allows users to manually add new personas through a structured form-based interface. Depending on the selected persona type, the form dynamically adjusts to capture relevant **Basic** and **Additional Information**. The purpose of this system is to ensure efficient categorization, documentation, and accessibility of individual profile data across departments.

4.2 Adding a Persona

To add a new persona:

- **Accessing the Dashboard**
Navigate to the Persona Module dashboard and click on the “**Add Manually**” button.
- **Filling Basic Information**
The basic fields are common across all persona types and include:
 - Name
 - Email
 - Phone number
 - State
 - PIN Code

- Message box
- **Filling Additional Information**
Based on the persona type selected, the module dynamically displays additional fields:
 - **Employees:** Date of Birth, Father's Name, Blood Group, Emergency Contact, Aadhar Number, Joining Date, Probation End Date, and Previous Employer.
 - **Vendors:** Address, PAN Number, GST Number, Bank Name, Account Number, IFSC Code.
 - **Customers:** Age, Location, Job, Income Range, Family Members, Weight, User Type, Wheelchair Type, Commute Range and Mode, Daily Pains, Commute Pains, Solutions Needed, Customer Segment, and Expected Gain.
 - **Others:** Limited to Basic Information only.

4.3 Saving the Persona

After filling in all required fields:

- Click the “**Save**” button to store the information.
- The newly created persona will be listed on the dashboard with its details and type.

4.4 Managing Personas

The dashboard provides powerful tools for ongoing persona management:

- **Edit:** Make changes to any existing persona.
- **Delete:** Remove a persona permanently from the system.
- **Mark as Favorite:** Highlight important personas for quick access.

The interface lists all personas with relevant details and includes **filter options** to sort by persona type, enabling quick and efficient navigation.

4.5 Notes and Best Practices

- Personas are **linked to backend data tables**, ensuring structured data storage.
- Deleting a persona will also **erase its associated data**, so caution is advised.
- Ensure **all required fields** are completed before saving to avoid errors.
- Regularly update persona data to reflect current information and maintain system reliability.

5. TECHNOLOGIES USED

The development of the Persona Module involved a modern tech stack focused on performance, scalability, and user experience. The key technologies and tools used include:

- **React.js:** Used for building the dynamic and responsive user interface, especially in managing components like forms, tabs, and theme switching.
- **Next.js:** Served as the framework for building the frontend with server-side rendering capabilities, improving load times and SEO.
- **Node.js:** Powered the backend, enabling efficient handling of API requests, data validation, and business logic.
- **JavaScript:** Formed the core scripting language for both frontend and backend logic.
- **PostgreSQL:** Used as the primary relational database to store all persona-related information securely, with structured schema support.

Key technical highlights:

- **Dynamic Form Handling:** Based on the selected persona type, the form updates in real time to display relevant fields, offering a personalized data entry experience.
- **Backend Integration:** Seamless interaction between frontend forms and the backend for saving, editing, deleting, and retrieving persona data.
- **State Management:** Efficient use of state hooks and context APIs to maintain and update UI elements instantly without page reloads.

6. CONCLUSION

The Persona Module project successfully delivers a robust and scalable solution for managing diverse user profiles. By integrating dynamic form handling with responsive UI components and a secure backend, the system efficiently categorizes and maintains personas, streamlining data management. The implementation of features such as customizable settings, detailed profile views, and comprehensive document handling showcases the effective use of modern web technologies like Node.js, React.js, Next.js, JavaScript, and PostgreSQL. Overall, the module demonstrates the potential to enhance operational efficiency and user experience in enterprise environments.